

Concurrent and Distributed Systems

Lab 6 – Testing Concurrent Programs

Materials

- **JUnit testing.pdf**
- Project archive: **cds.lab6.zip**, containing:
 - Calc, Account, Complex - simple sequential classes with corresponding test classes (CalcTest, AccountTest, ComplexTest)
 - MyBoundedBuffer – a concurrent program with its JUnit test (MyBoundedBufferTest)
- Further reading: **Java_9_Concurrency_Cookbook_Chapter_9_Testing_Concurrent_Applications.pdf**

Lab session

Activity 1 – Examples

1. Import the project **cds.lab6.zip** into NetBeans.
2. Examine the JUnit tests included in the project.
3. Run the tests and observe how they validate the behaviour of both sequential and concurrent classes.

Refer to JUnit testing.pdf and the lecture slides for explanations and guidance.

Activity 2 - Improving tests for MyBoundedBuffer

The class **MyBoundedBuffer** includes methods that are not covered by any test. Add new test cases that exercise these methods.

Activity 3 - Simple concurrent test of Racing Conditions

In Week 2, we examined an example of race condition in:

- **RaceCondition.java** (faulty implementation)
- **RaceCondition2.java** (correct implementation using synchronisation)

Write a concurrent JUnit test for each class:

- Use multiple threads running in parallel to increase the shared counter variable TOTAL.
- After all threads complete, use **assertEquals** to compare the final value of TOTAL to the expected result.

Expected outcomes:

- **RaceConditionTest** fails due to unsynchronised access.
- **RaceCondition2Test** passes because the increment method is synchronised.

Activity 4 - Concurrent tests for a Shared Account

Using the provided Account class as a starting point:

1. Implement a new class **SharedAccount** that uses synchronised methods to control access to shared data.
2. Write concurrent JUnit test cases for **SharedAccount**.

Activity 5 - Monitoring and Writing Log Messages

Read the following sections in

Java_9_Concurrency_Cookbook_Chapter_9_Testing_Concurrent_Applications.pdf and run the example programs presented:

- Monitoring a Lock interface
- Writing effective log messages

Resources

- J. Fernandez Gonzalez (2017), **Java 9 Concurrency Cookbook** (2nd edition), Packt Publishing. You can access the online version through the library (log in with your university account), link [here](#). The GitHub repository providing the complete Java classes for each book chapter example is here: <https://github.com/PacktPublishing/Java-9-Concurrency-Cookbook-Second-Edition>